

Step 4C — Fixed-Leader Visual HIL Comparison: Kuramoto and EAPF Consensus

MSc Project — BIOE70025

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1 Introduction

This Step 4C report documents a fixed-leader visual hardware-in-the-loop (HIL) comparison between two synchronisation models — Kuramoto and EAPF Consensus — under identical single-leader experimental conditions. It is **not** a final model-selection judgement; its purpose is to evaluate both models under the same visual closed-loop framework before moving to multi-neighbour mutual HIL testing.

Step 4A simulation-based model selection identified EAPF Consensus as the primary HIL candidate and retained Kuramoto as the secondary classical baseline [1, 2]. Step 4B showed that EAPF Consensus can be successfully transferred to the real Raspberry Pi visual loop. This Step 4C now compares both models side-by-side under the fixed-leader HIL setting.

2 Experimental Setup

- **Visual leader:** Laptop browser displaying a flashing white circle (350 px) on a black background at 2.0 Hz, 50 % duty cycle, controlled via REST API `/api/leader/config`.
- **Follower hardware:** Raspberry Pi 5 with Arducam 8 MP (IMX219) camera and GPIO 17 LED.
- **Flash detection:** Camera frames are processed through auto-ROI localisation, local contrast extraction, rolling adaptive normalisation, and hysteresis rising-edge detection with a 0.20 s min-interval gate.
- **Follower models:**
 - **Kuramoto** ($K = 3.5$): Leader phase is reconstructed from detected flash timestamps; coupling is $\dot{\theta}_F = \omega_F + K \cdot \sin(\hat{\theta}_L - \theta_F)$.
 - **EAPF Consensus** ($g_p = 0.02$, $g_f = 0.02$): Leader is treated as a single neighbour; consensus corrections use low-pass filtered, rate-limited phase and frequency updates from flash timestamps only.
- **Follower initial frequencies:** 1.5, 1.8, and 2.3 Hz.
- **Trial design:** 30 s per trial, 5 repeats per condition, 15 trials per model.
- **Synchronisation criterion:** Absolute nearest-flash timing error < 0.10 s sustained for 5 consecutive follower flash cycles.

3 Data Inclusion and Cleaning

The Kuramoto dataset uses the established 2026-06-11 batch (15 trials, 5 per condition). The EAPF dataset was collected on 2026-06-17. The Step4B.4 analysis originally included all valid directories in the output folder (39 total). After inspection of the terminal log, the formal batch was identified as the last 15 directories (time window 2026-06-17 18:25:17 to 18:33:50). The preceding 24 directories were preliminary smoke tests, alignment runs, and dry-runs and were excluded.

Table 1: Dataset summary.

Model	Source	Trials	Conditions	Reps	Status
Kuramoto	20260611_122117	15	3	5	Clean
EAPF	20260617_182517--183350	15	3	5	Clean

Two legitimate EAPF outliers were retained (not removed):

- 2.3 Hz repeat 4: time-to-sync ≈ 25.5 s (slow convergence).
- 2.3 Hz repeat 5: steady-state MAE ≈ 0.107 s (borderline above the 0.10 s threshold).

These reflect real visual-loop timing variability rather than data errors and are visible as individual points in Figures 2 and 3.

4 Results

Table 2: Fixed-leader visual HIL — summary by condition. Ranges show minimum to maximum across 5 repeats.

Model	Condition	Sync	TTS (s)	MAE (s)	Leader FCR	Loop (Hz)
Kuramoto	1.5 Hz	100 %	2.8–4.3	0.043–0.070	0.95–1.00	—
Kuramoto	1.8 Hz	100 %	2.6–3.4	0.000–0.006	0.98	—
Kuramoto	2.3 Hz	100 %	3.3–5.5	0.052–0.063	0.95–1.00	—
EAPF	1.5 Hz	100 %	7.2–10.0	0.000–0.091	0.93–0.95	11.2–12.1
EAPF	1.8 Hz	100 %	3.9–10.0	0.046–0.062	0.93–0.95	10.7–10.8
EAPF	2.3 Hz	100 %	6.8–25.5	0.044–0.107	0.95	10.7–12.5

Both models achieved 100 % synchronisation success under the tested single-leader visual HIL conditions. Detailed comparison figures follow.

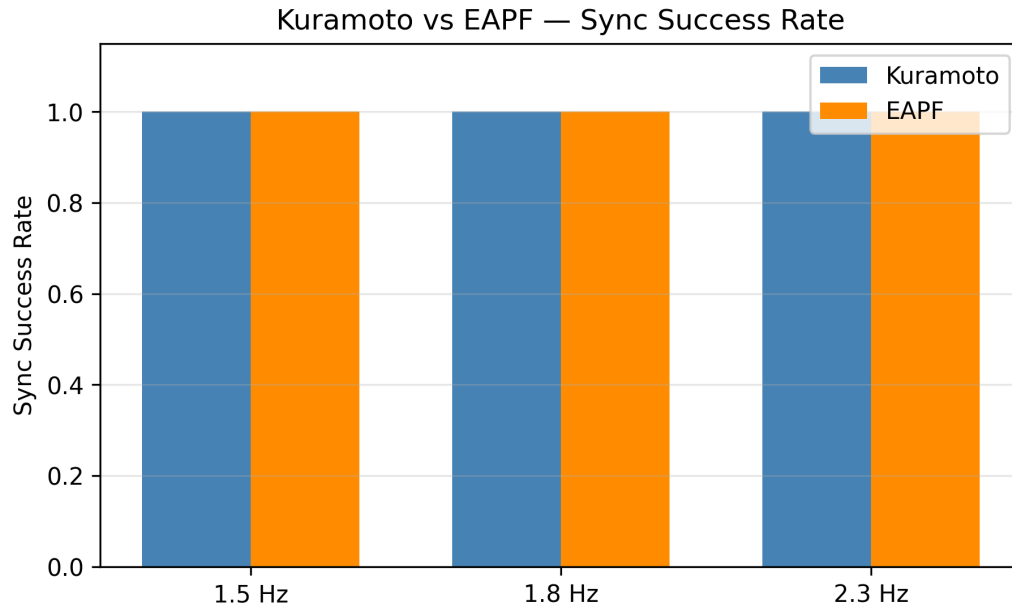


Figure 1: Synchronisation success rate by model and condition. Both models achieve 100 % across all follower initial frequencies.

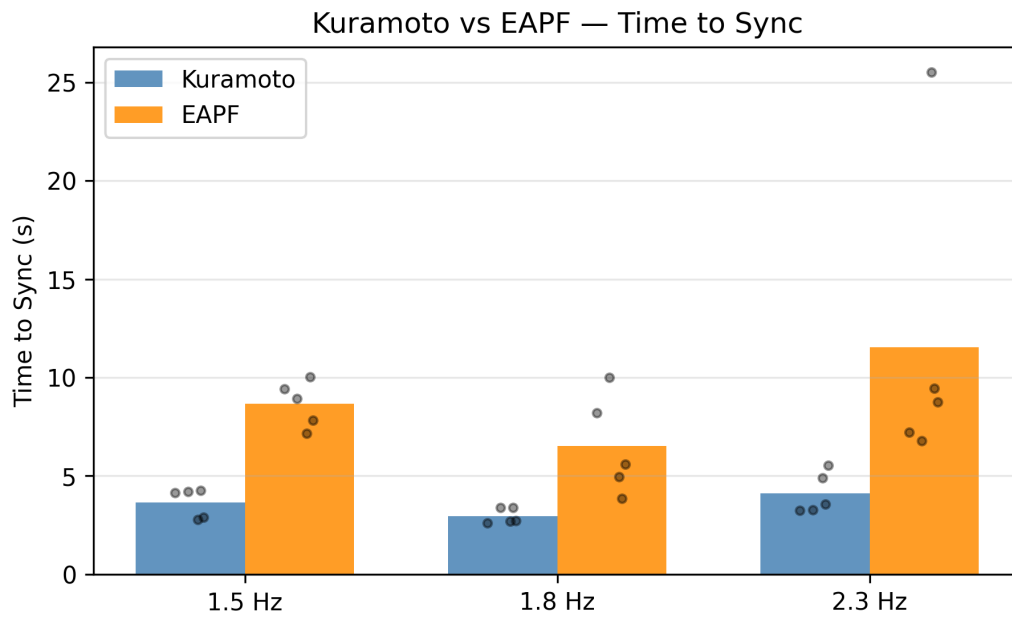


Figure 2: Mean time to synchronisation by model and condition. Bars show mean; individual trial points are overlaid in black. Kuramoto converges faster (2.6–5.5s) than EAPF (3.9–25.5s). The EAPF 2.3 Hz bar is inflated by a single 25.5s outlier, visible as the isolated black point above the bar.

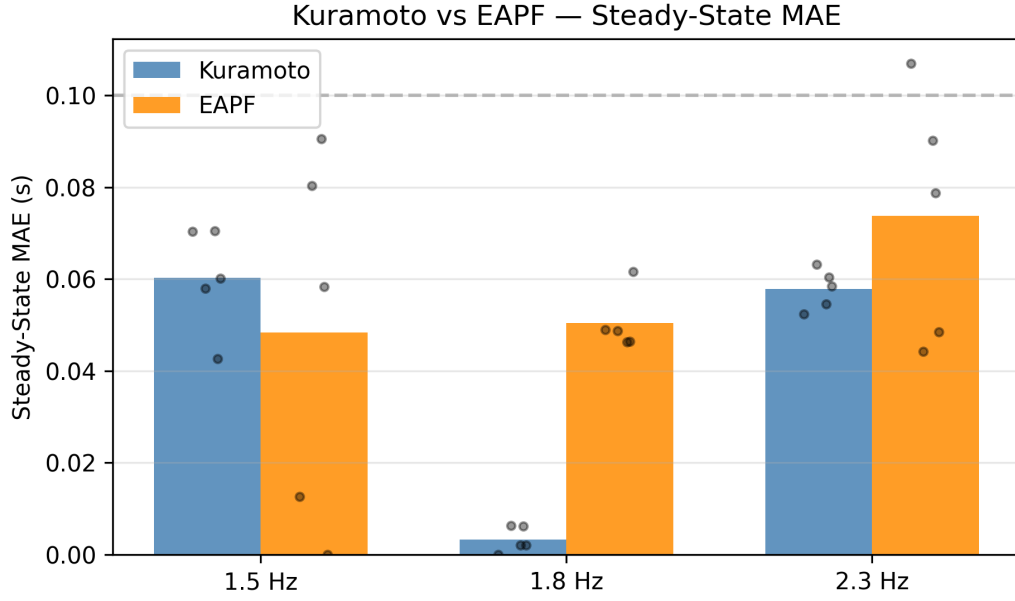


Figure 3: Steady-state mean absolute timing error by model and condition. Bars show mean; individual trial points overlaid in black. Dashed line at 0.10 s marks the synchronisation threshold. Both models generally remain below or close to 0.10 s; one retained EAPF 2.3 Hz trial is borderline above at ≈ 0.107 s, visible as the isolated point.

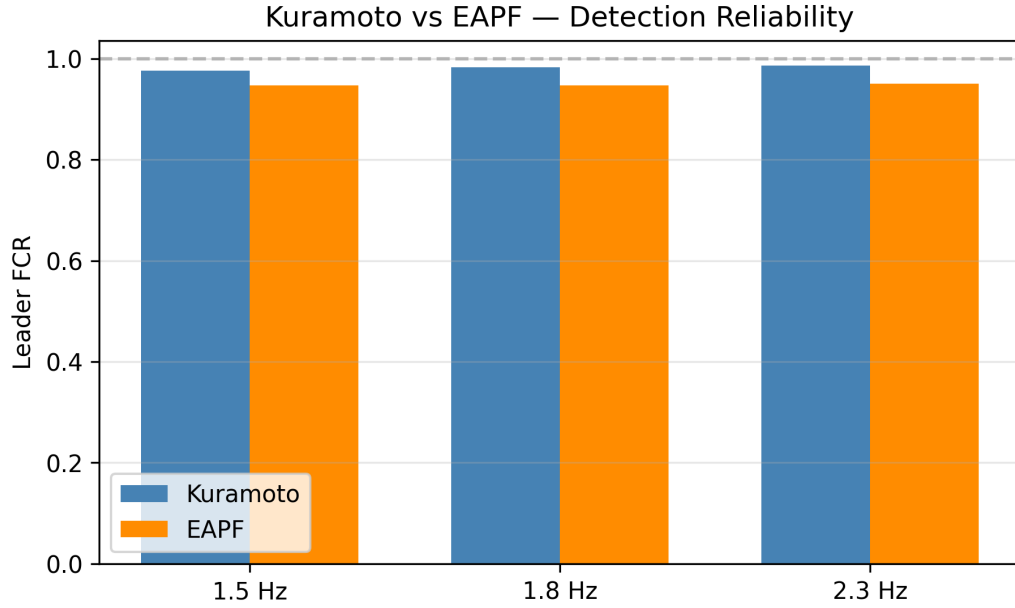


Figure 4: Leader flash detection reliability (leader flash count ratio) by model and condition. Dashed line at 1.0 marks perfect detection. Both models operate with comparable reliability (0.93–1.00), confirming that detection performance is dominated by the camera pipeline, not model differences.

5 Discussion

Successful transfer. Both Kuramoto and EAPF Consensus achieved 100% synchronisation success in the fixed-leader visual HIL setting. This confirms that EAPF Consensus, designed for

multi-neighbour consensus, also functions robustly as a single-leader follower using only camera-detected flash timestamps.

Time to synchronisation. Kuramoto converges faster (2.6–5.5 s) than EAPF Consensus (3.9–10.0 s typical, one 25.5 s outlier). This is consistent with the Kuramoto model’s continuous coupling versus EAPF’s slow consensus gains (0.02). For applications requiring rapid lock, Kuramoto has an advantage; for robust distributed synchronisation, EAPF’s convergence profile is acceptable within a 30 s window.

Steady-state accuracy. Both models generally remain below or close to the 0.10 s synchronisation threshold. Kuramoto MAE range is 0.000–0.070 s; EAPF MAE range is 0.000–0.107 s. The EAPF upper bound is driven by a single retained 2.3 Hz outlier trial (MAE $\approx$ 0.107 s) and does not indicate a systematic accuracy problem.

Detection reliability. Leader FCR is comparable across models (0.93–1.00 for Kuramoto, 0.93–0.95 for EAPF). This confirms that the shared camera detection pipeline, not model-specific behaviour, is the primary determinant of detection reliability.

Step 4C does not contradict Step 4A. The faster Kuramoto convergence in this fixed-leader test does not contradict the Step 4A model-selection result. Step 4A selected EAPF Consensus because of its stronger multi-neighbour and N=5 scalability performance (100 % zero-lag group sync at N=3 and N=5 in simulation), whereas Step 4C only evaluates single-follower entrainment to a fixed visual leader. The two stages answer different questions: Step 4A asks “which model synchronises best in a decentralised multi-agent network,” while Step 4C asks “can both models function under real camera-based visual detection.” The answer to both is affirmative.

Outlier handling. The two EAPF 2.3 Hz outliers (slow sync, borderline MAE) are retained as real trials. They likely reflect the challenge of converging from a faster initial frequency (2.3 Hz) toward a slower leader (2.0 Hz) with very low consensus gains, combined with natural visual-loop timing variability. They are not excluded.

Not the final comparison. This fixed-leader HIL comparison should not be interpreted as a final model-selection verdict. The more important next validation is Step 5 mixed-reality mutual / multi-neighbour HIL, where EAPF Consensus is expected to demonstrate the scalability advantages observed in Stage 4A simulation.

6 Limitations

- The fixed-leader setup evaluates single-follower tracking, not decentralised mutual synchronisation for which EAPF Consensus was designed.
- Only one real Pi follower is used; the leader is a browser simulation.
- Camera detection reliability, loop rate, ambient lighting, screen brightness, and alignment affect results across trials.
- EAPF and Kuramoto may behave very differently in mutual multi-neighbour settings — the Step 4A simulation results suggest EAPF Consensus scales better to larger agent counts.
- Kuramoto uses a different, lower coupling gain in the single-leader HIL test ($K = 3.5$) than the value locked for Step 4A multi-neighbour simulation ($K = 5.0$). Therefore, the fixed-leader HIL timing results should not be interpreted as a direct validation of the Step 4A multi-neighbour Kuramoto setting.

- Flight testing and coordinated action triggering are not part of Step 4C.

7 Conclusion and Next Step

Step 4C confirms that both Kuramoto and EAPF Consensus can be evaluated in the same fixed-leader visual HIL framework with 100 % synchronisation success under the tested conditions. Kuramoto provides a strong classical single-leader baseline; EAPF Consensus successfully transfers its flash-event-driven consensus algorithm to the real Pi visual loop.

Step 5 will proceed directly to mixed-reality mutual HIL, starting with 1 virtual agent + 1 real Pi agent mutual convergence, then extending to 2 virtual agents + 1 real Pi all-to-all / multi-neighbour HIL.

References

- [1] Y. Kuramoto, “Self-entrainment of a population of coupled non-linear oscillators,” in *Int. Symp. on Mathematical Problems in Theoretical Physics*, Springer, 1975, pp. 420–422.
- [2] F. M. Gardner, *Phaselock Techniques*, 3rd ed. Hoboken, NJ: John Wiley & Sons, 2005.
- [3] R. E. Mirollo and S. H. Strogatz, “Synchronization of pulse-coupled biological oscillators,” *SIAM J. Appl. Math.*, vol. 50, no. 6, pp. 1645–1662, 1990.
- [4] R. Olfati-Saber, J. A. Fax, and R. M. Murray, “Consensus and cooperation in networked multi-agent systems,” *Proc. IEEE*, vol. 95, no. 1, pp. 215–233, 2007.